

Fiscal year-end close — design proposal

This document describes a best-practice scheme for closing financial books at a configurable fiscal year-end, optionally purging ledger detail and other historical tables, while preserving balances, arrears, active loans, and collection-engine accuracy.

It aligns with FundFlow’s existing model:

- **Pool mirrors** — master cash/fund = sum of member cash/fund (`MasterAccountInvariantService`, `MemberInvariantService`)
- **Contribution cycles** — open period, arrears, late fees (`ContributionCycleService`, `ContributionCollectionCycleService`)
- **Incremental collection** — cash-in triggers contribution and EMI collection
- **Import cut-off pattern** — `MemberOpeningBalanceService`, `contribution_arrears_cutoff_date`, `opening_*_balance` on members

Status: Phases 1–4 implemented (readiness, snapshot, roll-forward, books lock, Tier A/B purge, archive exports). Phase 5 (multi-year history / restore tooling) is optional follow-up.

1. Core principle: separate “state” from “history”

Layer	Purpose after close	Purge candidate?
Balances (<code>accounts.balance</code> , master pool totals)	Current financial position	Never — roll forward
Operational carry-forward (unpaid contributions, pending/overdue installments, active loans)	What the collection engine must still act on	Never without re-encoding
Ledger detail (<code>transactions</code> , bank lines, audit rows)	Proof / drill-down	Yes, after snapshot
Closed workflow (accepted deposits, paid installments, posted contributions)	Historical narrative	Yes, if summarized in close snapshot

Rule: Purge only after a **certified close snapshot** captures everything needed to recompute invariants without the deleted rows.

`MemberInvariantService` defines expected balances as:

`expected = opening_* +  $\Sigma$ (transactions by reference type)`

Closing must either:

1. **Roll opening balances forward** and purge transactions (invariant still works), or

2. **Keep minimal synthetic opening legs** (one tagged entry per pool per member) and purge the rest. Rolling forward on `members.opening_*` matches the existing CSV import cut-off pattern.

2. Configuration

Tenant settings (proposed: **Settings** → **Fiscal calendar**):

Setting key	Example	Use
<code>fiscal_year_start_month</code>	1 (Jan) or 7 (Jul)	Fiscal year boundary
<code>fiscal_year_start_day</code>	1	Optional; default 1st
<code>books_closed_through</code>	2025-12-31	Last date in closed books; immutable once closed
<code>current_fiscal_year_label</code>	FY2025	Display / reporting
<code>purge_policy</code>	<code>archive_then_delete / delete_only / retain_7y</code>	Post-close data retention

Business day (`BusinessDaySettings`) drives operational “today” for collection. **books_closed_through** is a hard ceiling for backdated postings and for distinguishing historical vs current periods.

3. Close workflow

```
stateDiagram-v2
    [*] --> OpenYear
    OpenYear --> PreCloseValidation: Admin starts close
    PreCloseValidation --> Blocked: Gates fail
    Blocked --> PreCloseValidation: Remediate
    PreCloseValidation --> SnapshotBuilt: All gates pass
    SnapshotBuilt --> PendingApproval: Admin review
    PendingApproval --> RolledForward: Approve and execute
    RolledForward --> Purged: Optional purge job
    Purged --> Locked: books_closed_through set
    Locked --> OpenYear: New fiscal year operations
```

Phase A — Pre-close validation (hard gates)

Block close unless all pass (or explicit waiver with audit):

1. **Pool mirrors** — `MASTER_CASH_POOL_DRIFT`, `MASTER_FUND_POOL_DRIFT` within tolerance
2. **Member drift** — no `MEMBER_CASH_DRIFT` / `MEMBER_FUND_DRIFT` for active members
3. **Open operational items** — no pending fund postings, cash-outs, or in-flight loan disbursements (configurable)
4. **Reconciliation** — no open critical reconciliation exceptions; uncleared bank lines documented or matched
5. **Contribution cycles** — every cycle close date is posted/collected or explicitly carried as arrears
6. **Loans** — active loans consistent: `amount_disbursed`, `total_repaid`, installment totals vs fund/cash legs

Output: Close Readiness Report (in-app checklist + exportable PDF/CSV).

Phase B — Snapshot (immutable anchor)

Capture tenant-wide and per-member state before any purge. See §6 Proposed schema.

Phase C — Roll-forward (preserves standing)

1. **Per member** — set `opening_cash_balance` / `opening_fund_balance` to closing balances; update `opening_balances_posted_at`
2. **Optional** — post one tagged leg pair per pool (`FISCAL_CLOSE_FY2025`) via `MemberOpeningBalanceService`-style posting, or rely on balance + opening fields only if all transactions will be purged
3. **Contributions** — retain unpaid/partial/overdue rows; archive or delete fully settled rows for periods `period_end` only after snapshot
4. **Loans** — retain active loans and unpaid/overdue installments; purge paid installment history only after snapshot confirms `total_repaid` and schedule state
5. **Collection engine** — extend import cut-off guards with tenant `books_closed_through` so auto-collection never targets purged periods

Phase D — Purge (optional, policy-driven)

Background job only after roll-forward and export complete.

Tier	Data	Condition
A	<code>transactions</code> (<code>period_end</code>), cleared <code>bank_transactions</code> , resolved <code>reconciliation_exceptions</code> , old notifications	After snapshot
B	Posted <code>contributions</code> , paid <code>loan_installments</code> , closed <code>fund_postings</code> , <code>fund_audit_log</code> , monthly statements	Export first (default)
C	<code>members</code> , <code>accounts</code> , open arrears, active <code>loans</code> + open installments, <code>fiscal_closes</code> + snapshots, settings	Never

Bank caveat: Do not purge uncleared bank lines tied to pending cash-outs or deposits unless explicitly written off during close.

Phase E — Lock

- Reject ledger postings with `transacted_at` `books_closed_through` (except controlled super-admin adjustment workflow)
- UI banner when books are closed and/or ledger purged
- New fiscal year runs on clean transaction table + rolled opening balances + carried arrears

4. Financial standing after close

Question	Source after purge
Member cash / fund balance	<code>accounts.balance</code> (+ invariant vs <code>opening_*</code>)
Who owes contributions?	Remaining <code>contributions</code> rows + snapshot <code>contribution_arrears_json</code>
Loan outstanding and next EMI	Active <code>loans</code> + pending/overdue <code>loan_installments</code>
Delinquency / eligibility	Evaluators on carried cycles; retain counters on <code>members/loans</code> if history purged
Master pool = Σ members	Unchanged balances; nightly reconciliation
“What happened in FY2025?”	<code>fiscal_closes</code> export / archived GL, not live tables

5. Alignment with existing code (reuse, don’t reinvent)

Fiscal close should extend patterns already in production—not introduce a parallel accounting model.

Existing piece	Location	Fiscal close reuse
Opening balance posting	<code>MemberOpeningBalanceService</code>	Roll forward legs and <code>opening_cash_balance /</code> <code>opening_fund_balance /</code> <code>opening_balances_posted_at</code> ; pass <code>entry_label</code> such as <code>FISCAL_CLOSE_FY2025</code> (same as <code>IMPORT_CUTOFF</code> today)
Per-member arrears cut-off	<code>members.contribution_arrears</code>	“Doesn’t cut off by date”; for fully settled members at close, align with <code>period_end</code> ; complement with tenant-level <code>books_closed_through</code>
Pre-cutoff contribution dismissal	<code>ContributionCollectionService</code>	Safe reversal of pending rows before a cut-off (<code>Dismissed: before</code> <code>contribution arrears cut-off</code>); model for excising pre-close rows without breaking collection state
Member drift formula	<code>MemberInvariantService</code>	Post-close regression: <code>expected balances =</code> <code>opening_* + Σ(transactions since close)</code> ; must pass after roll-forward and purge
Master pool drift	<code>MasterAccountInvariantService</code>	Reconcile; before close; re-run after purge
Pool mirror posting	<code>AccountingService</code> (<code>creditMemberCashWithMasterMirror</code> etc.)	Same mirror rules when posting optional legs; use <code>withoutMemberCashCollection()</code> during roll-forward
Import cut-off on approval	<code>MembershipApprovalPostProcessor</code>	“Ping Pong line”; system takes over at date X with balances Y”
Business day override	<code>BusinessDay</code> , <code>BusinessDaySettings</code>	Close <code>execution</code> timestamp; fiscal <code>period_end</code> is configured calendar, not necessarily “today”


```

purge_summary_json      json nullable -- row counts deleted per table
failure_reason          text nullable
created_at, updated_at

```

Indexes: status, period_end, unique partial on fiscal_year_label where status not in (failed).

6.3 fiscal_close_member_snapshots

One row per member per close.

fiscal_close_member_snapshots

```

id                bigint PK
fiscal_close_id   FK fiscal_closes
member_id         FK members
cash_balance      decimal(15,2)
fund_balance      decimal(15,2)
opening_cash_before decimal(15,2) nullable
opening_fund_before decimal(15,2) nullable
contribution_arrears_json json          -- [{period, principal_due, late_fee_due, collection_status, ...}]
loans_json        json          -- [{loan_id, outstanding, overdue_installments[], next_due_date, ...}]
delinquency_json  json nullable -- trailing/rolling misses at close
eligibility_json  json nullable -- flags for audit
created_at, updated_at

```

UNIQUE (fiscal_close_id, member_id)

6.4 fiscal_close_waivers (optional)

Explicit override when a gate fails but admin proceeds.

fiscal_close_waivers

```

id                bigint PK
fiscal_close_id   FK fiscal_closes
gate_code         varchar      -- e.g. RECON_EXCEPTION_OPEN
reason           text
waived_by         FK users
created_at

```

6.5 Existing tables — fields reused (no migration required for v1)

Table / field	Role in close
members.opening_cash_balance	Roll-forward target
members.opening_fund_balance	Roll-forward target
members.opening_balances_posted_at	Close execution timestamp
members.contribution_arrears_cutoff_date	Per-member “do not collect before”; may align with period_end for settled members
accounts.balance	Source of truth post-close

6.6 Optional v2: fiscal_close_archive_refs

Pointer to cold storage (S3) for GL CSV, PDF packs.

fiscal_close_archive_refs

```

id                bigint PK
fiscal_close_id   FK fiscal_closes

```

kind	varchar	-- gl_export/arrears_aging/loan_portfolio/readiness_report
storage_disk	varchar	
storage_path	varchar	
byte_size	bigint nullable	
created_at		

7. Service outline

7.1 Package layout

```
app/
  Services/FiscalClose/
    FiscalCloseService.php           -- orchestrator
    FiscalCloseReadinessService.php  -- Phase A gates
    FiscalCloseSnapshotService.php   -- Phase B
    FiscalCloseRollForwardService.php -- Phase C
    FiscalClosePurgeService.php       -- Phase D
    FiscalCloseExportService.php      -- GL / arrears / loan exports
    FiscalClosePeriodResolver.php     -- FY boundaries from settings
  Models/Tenant/
    FiscalClose.php
    FiscalCloseMemberSnapshot.php
    FiscalCloseWaiver.php
  Filament/Tenant/Pages/
    FiscalYearClose.php             -- admin wizard
```

7.2 FiscalClosePeriodResolver

```
// Responsibilities
resolveCurrentFiscalYear(): FiscalYearDefinition
resolvePeriodForLabel(string $label): { start: Carbon, end: Carbon }
nextPeriodAfter(Carbon $closedThrough): FiscalYearDefinition
assertNotClosed(Carbon $transactedAt): void // throws if <= books_closed_through
```

Uses `Setting::get('fiscal', ...)` and `BusinessDay` where operational “today” matters.

7.3 FiscalCloseReadinessService

```
assess(?Carbon $proposedPeriodEnd = null): FiscalCloseReadinessReport
```

```
// Gates (each returns Pass/Fail/Warn + details)
checkMasterPoolMirrors(): GateResult
checkMemberDrifts(): GateResult
checkOpenFundPostings(): GateResult
checkOpenCashOuts(): GateResult
checkReconciliationExceptions(): GateResult
checkUnclearedBankLines(): GateResult
checkContributionCycleCompleteness(Carbon $periodEnd): GateResult
checkLoanPortfolioConsistency(): GateResult
```

```
canProceed(FiscalCloseReadinessReport $report, Collection $waivers): bool
```

Delegates to existing `ReconciliationService`, `MasterAccountInvariantService`, `MemberInvariantService`.

7.4 FiscalCloseSnapshotService

```
build(FiscalClose $close): FiscalClose
```

```
capturePoolTotals(): array
```

```
captureMemberSnapshot(Member $member, Carbon $periodEnd): FiscalCloseMemberSnapshot
```

```
computeChecksum(Collection $snapshots): string
```

Populates `fiscal_closes.pool_snapshot_json` and `fiscal_close_member_snapshots.*`.

7.5 FiscalCloseRollForwardService

```
execute(FiscalClose $close): void
```

```
// Per member (transactional)
```

```
rollMemberOpeningBalances(Member $member, FiscalCloseMemberSnapshot $snapshot): void
```

```
// Delegates to MemberOpeningBalanceService with entry_label FISCAL_CLOSE_{label}
```

```
// Sets opening_* from snapshot; does not change accounts.balance
```

```
applyTenantBooksClosedThrough(Carbon $periodEnd): void
```

```
// Setting::set('fiscal', 'books_closed_through', ...)
```

```
pruneSettledContributions(FiscalClose $close, PurgePolicy $policy): int // optional in Phase C or D
```

```
prunePaidInstallments(FiscalClose $close, PurgePolicy $policy): int
```

Important: Reuse `AccountingService::withoutMemberCashCollection()` during roll-forward posting to avoid auto-collection side effects.

7.6 FiscalClosePurgeService

```
execute(FiscalClose $close, PurgePolicy $policy): PurgeSummary
```

```
purgeTransactionsThrough(Carbon $periodEnd): int
```

```
purgeClearedBankLinesThrough(Carbon $periodEnd): int
```

```
purgeResolvedReconciliationExceptions(): int
```

```
purgePostedContributionsThrough(Carbon $periodEnd): int // Tier B
```

```
purgePaidLoanInstallmentsThrough(Carbon $periodEnd): int // Tier B
```

```
// Never: members, accounts, open contributions, open installments, fiscal_closes
```

Runs in queued job with chunking; updates `fiscal_closes.purge_summary_json`.

Post-purge: run `MasterAccountInvariantService` + sample `MemberInvariantService` checks.

7.7 FiscalCloseService (orchestrator)

```
startDraft(string $fiscalYearLabel): FiscalClose
```

```
runValidation(FiscalClose $close): FiscalCloseReadinessReport
```

```
buildSnapshot(FiscalClose $close): FiscalClose
```

```
submitForApproval(FiscalClose $close): FiscalClose
```

```
approveAndRollForward(FiscalClose $close, User $approver): FiscalClose
```

```
schedulePurge(FiscalClose $close): void
```

```
cancel(FiscalClose $close): void // only if not rolled_forward
```

State machine enforced on `fiscal_closes.status`.

7.8 FiscalCloseExportService

```
exportG1(FiscalClose $close): string           // storage path
exportArrearsAging(FiscalClose $close): string
exportLoanPortfolio(FiscalClose $close): string
exportReadinessReport(FiscalClose $close): string
```

Called after snapshot, before purge.

7.9 Integration hooks (existing services)

Service	Change
AccountingService::debit/credit	Call <code>FiscalClosePeriodResolver::assertNotClosed(\$transactedA</code> when set
ContributionCollectionCycleService	Skip periods <code>books_closed_through</code> (tenant) in addition to member cut-off
LoanInstallmentCollectionService	Same period ceiling for auto-collection
MemberOpeningBalanceService	Accept <code>entry_label</code> e.g. <code>FISCAL_CLOSE_FY2025</code> (already supports label param)
FundAuditLogService	Log <code>FISCAL_CLOSE_*</code> events

8. Admin UX (high level)

8.1 Settings — Fiscal calendar

Path: Tenant admin → **Settings** → **General** (or dedicated **Fiscal calendar** section)

Control	Purpose
Fiscal year start month / day	Defines FY boundaries for labels and readiness reports
Current fiscal year label	Display (e.g. <code>FY2025</code>)
Books closed through	Read-only once set; shows last closed period
Purge policy	Default retention behavior after close

Optional: link to **Business calendar** (`BusinessDaySettings`) with helper text explaining the difference—business day drives collection “today”; fiscal calendar drives book lock and purge scope.

8.2 Year-end close wizard

Path: Tenant admin → **Accounting** → **Year-end close** (new Filament page: `FiscalYearClose`)

Step	Admin action	System behavior
1	Select fiscal year / confirm <code>period_end</code>	<code>FiscalClosePeriodResolver</code> computes inclusive start/end
2	Run readiness checks	Live checklist from <code>FiscalCloseReadinessService</code> ; failures link to reconciliation, deposits, cash-outs, etc.
3	Preview snapshot	Compare pool totals and sample members vs live <code>accounts.balance</code>

Step	Admin action	System behavior
4	Download exports	GL CSV, arrears aging, loan portfolio, readiness PDF via <code>FiscalCloseExportService</code>
5	Approve	Optional second approver; record <code>approved_by</code> / <code>approved_at</code>
6	Execute roll-forward	<code>FiscalCloseRollForwardService</code> ; set <code>books_closed_through</code>
7	Schedule purge (optional)	Confirm tier A/B per <code>purge_policy</code> ; queued job with progress

UX patterns: Match existing operational pages (contributions apply, reconciliation)—confirmation modals for irreversible steps, disabled purge until exports exist, clear “blocked” state with waiver flow for exceptional gates.

8.3 Post-close visibility

- **Admin / member panel banner** when `books_closed_through` is set (similar to `partials/business-day-banner.bl` e.g. “Books closed through 31 Dec 2025; historical ledger archived.”)
- **Close history** — read-only list of `fiscal_closes` with status, checksum, export download links, purge summary.
- **Member profile** — optional read-only “Opening balances (FY close)” showing rolled `opening_*` and close date (no member-facing purge).

8.4 Audit trail

Every state transition writes to:

- `fiscal_closes` status fields and `failure_reason`
- `FundAuditLogService` — events such as `FISCAL_CLOSE_VALIDATED`, `FISCAL_CLOSE_SNAPSHOT`, `FISCAL_CLOSE_ROLLED_FORWARD`, `FISCAL_CLOSE_PURGE_COMPLETED`
- `fiscal_close_waivers` when gates are overridden with documented reason

Purge job logs **row counts per table** in `purge_summary_json` for forensic review.

9. Testing strategy

Test	Assert
Readiness blocks on pool drift	Close cannot proceed
Snapshot matches live balances	Checksum stable
Roll-forward + purge Tier A	<code>MemberInvariantService</code> passes with opening + new-year txs only
Arrears retained	Unpaid contribution still collectable after close
Open loan EMI	Pending installment still collectable
Backdated posting rejected	<code>transacted_at <= books_closed_through</code> fails
Purge does not delete open bank uncleared	Gate or explicit write-off required

Feature tests under `tests/Feature/Tenant/FiscalClose/` using `InitializesTenancy`.

10. Implementation phases (when you're ready)

Implement incrementally. Do **not** ship purge until roll-forward and invariants are proven in production-like tests.

Phase	Deliverable	Outcome	Risk
1 — Readiness only	Fiscal settings; <code>FiscalCloseReadinessService</code> ; Close Readiness Report UI; no mutations	Admins can see whether the tenant <i>could</i> close safely	Low
2 — Snapshot + roll-forward	Migrations for <code>fiscal_closes</code> / snapshots; <code>FiscalCloseSnapshotService</code> , <code>FiscalCloseRollForwardService</code> ; <code>books_closed_through</code> lock in <code>AccountingService</code>	Certified close with opening roll-forward; ledger retained	Medium
3 — Purge Tier A	<code>FiscalClosePurgeService</code> for <code>transactions</code> , cleared bank lines, resolved exceptions; invariant tests post-purge	Smaller DB; live balances unchanged	High
4 — Purge Tier B + exports	Archive posted contributions, paid installments, closed fund postings; mandatory export manifest	Full year-end hygiene with offline archive	High
5 — History & restore (optional)	Multi-year <code>fiscal_closes</code> list; documented restore-from-export procedure (manual, not auto-replay)	Long-term compliance and audit	Optional

Suggested order of work

1. Phase 1 — unblocks operational review without risk.
2. Phase 2 — first “real” close on a staging tenant; verify `MemberInvariantService` / `MasterAccountInvariantService` before and after.
3. Phase 3 — only after at least one successful close with exports stored.
4. Phase 4 — only if storage/retention policy requires deleting Tier B rows.

Exit criteria per phase

- **Phase 1:** All readiness gates runnable; report exportable; zero data mutation.
- **Phase 2:** Roll-forward idempotent; backdated posting blocked; arrears and open EMIs still collectable.
- **Phase 3:** Post-purge pool mirrors hold; sample of members pass drift check with `opening_*` + new-year txs only.
- **Phase 4:** Export manifest complete before any Tier B delete; purge summary auditable.

11. Risks and explicit non-goals

11.1 Risks

Risk	Impact	Mitigation
Purge before snapshot	Unrecoverable loss of proof; broken invariants	State machine: snapshot required before roll-forward; exports before purge
Delete unpaid contribution rows	Collection engine “forgets” arrears	Tier C never purged; snapshot <code>contribution_arrears_json</code> ; keep pending/overdue rows

Risk	Impact	Mitigation
Delete unpaid / overdue installments	EMI auto-collection targets wrong periods	Keep open installments; snapshot <code>loans_json</code>
Purge uncleared bank lines	Breaks deposit/cash-out clearance	Readiness gate; explicit write-off workflow
Purge paid installment history	Loan eligibility / late-payment counters wrong	Snapshot delinquency/eligibility; retain <code>late_repayment_count</code> on <code>loans/members</code> or encode in snapshot
Roll-forward triggers auto-collection	Unexpected debits on close day	<code>AccountingService::withoutMemberCashCollection()</code> during roll-forward
FY boundary vs contribution cycle	Confusion on which periods are “open”	Readiness report documents overlap; tenant <code>books_closed_through</code> + existing open-period logic
Hard delete vs archive	Compliance failure	Default <code>purge_policy = archive_then_delete</code>

11.2 Anti-patterns (do not do)

- **Big bang purge** — deleting transactions without per-member opening roll-forward.
- **Balance reset** — zeroing `accounts.balance` and reposting everything; causes visible jumps and breaks bank clearance links.
- **Regenerating loan schedules** after close — carry forward remaining installments only.
- **Silent close** — no export, no checksum, no audit log entries.
- **Member-level close only** — fiscal close is tenant-wide; per-member cut-off remains a separate concept (`contribution_arrears_cutoff_date`).

11.3 Explicit non-goals (v1)

Non-goal	Notes
Full double-entry GL with retained journal numbers across years	Exports provide external GL; live table is operational sub-ledger
Automatic unattended close	Always requires admin validation and approval
Cross-tenant consolidation	Single-tenant scope only
Auto-replay from export into live ledger	Restore is manual / forensic, not production workflow
Rewriting historical contribution cycle stubs	Use carry-forward rows + snapshot, not regeneration from ledger
Purging active members, accounts, or open loans	Tier C forbidden

12. Recommendation

Treat **fiscal year-end close as a tenant-wide import cut-off**—the same conceptual move as CSV migration (`MemberOpeningBalanceService`, `contribution_arrears_cutoff_date`), applied once per fiscal year for the whole fund.

Recommended close sequence

1. **Validate** — pool and member invariants; no blocking operational items (or documented waivers).
2. **Snapshot** — balances, arrears, loan state, checksum; export GL and reports.

3. **Roll forward** — update `opening_*` on each member (and optionally one tagged `FISCAL_CLOSE_*` mirrored leg per pool); set `books_closed_through`.
4. **Purge** — ledger detail only, under policy, never operational carry-forward (Tier C).
5. **Lock** — reject backdated postings; new fiscal year runs on a clean transaction spine plus rolled openings and carried arrears.

This yields a smaller database and faster ledgers while keeping contributions, loans, repayments, arrears, delinquency, and eligibility behavior consistent with how the application already models money.

Default policy choices (see §13 for alternatives)

- Fiscal year: **calendar month/day** (Phase 1).
- Purge: **Standard** — Tier A + B with mandatory `archive_then_delete`.
- Opening legs: **One leg per pool per member** via `MemberOpeningBalanceService`; optional balance-only mode for tenants that do not need residual transaction rows.

When implementation starts, confirm §13 explicitly; then build Phase 1 readiness before any mutating close logic.

13. Policy decisions required before implementation

Confirm these three choices to finalize migrations and service contracts:

13.1 Fiscal year definition

Option	Description
A — Calendar	<code>fiscal_year_start_month / day</code> (e.g. 1 Jan – 31 Dec)
B — Contribution-aligned	FY boundary aligned to <code>contribution.cycle_start_day</code>
C — Hybrid	Calendar label, but collection “open period” may span FY boundary with explicit rules

Recommendation: Start with **A**; document overlap with contribution cycles in readiness report.

13.2 Purge aggressiveness

Option	Description
Conservative	Tier A only (transactions + cleared bank); retain posted contributions and paid installments
Standard	Tier A + B with mandatory export archive
Aggressive	Tier B delete without archive (not recommended for production)

Recommendation: **Standard** with `archive_then_delete` default.

13.3 Synthetic opening legs

Option	Description
Balance-only	Update <code>opening_*</code> and purge all transactions; invariant uses opening + post-close txs only
One leg per pool	Post single <code>FISCAL_CLOSE_*</code> mirrored leg per member via <code>MemberOpeningBalanceService</code> ; purge rest

Option	Description
Full retain opening legs	Keep all opening/cut-off legs forever; purge only non-opening transactions

Recommendation: One leg per pool for easier debugging and alignment with `MemberInvariantService` component sums; optional **balance-only** mode for tenants that never drill into ledger detail.

14. Reference implementations in codebase

See also §5 [Alignment with existing code](#) for the full reuse matrix.

Flow	Service / doc	Pattern
CSV import cut-off balances	<code>MemberOpeningBalanceService</code>	Signed pool mirrors on member + master
Arrears cut-off	<code>ContributionCollectionCycleService</code>	Skip / dismiss pre-cutoff periods
Member drift formula	<code>MemberInvariantService</code>	<code>opening_*</code> + transaction components
Master pool drift	<code>MasterAccountInvariantService</code>	Nightly batch
Migration stubs	<code>docs/fund_management_system_requirements.md §4</code>	Historical cycle handling — analogous to fiscal carry-forward

15. Should we proceed with implementation?

Yes — but incrementally, not as one big release.

15.1 Recommendation

Proceed with Phase 1 only (fiscal settings + readiness report + wizard shell, **no mutations, no purge**). That is low risk, validates the design against real tenant data, and forces resolution of open policy questions (§13) before anything irreversible.

Do not proceed yet with Phases 2–4 (roll-forward, lock, purge) until:

1. **Policy choices in §13 are confirmed** — especially FY definition, purge tier, and synthetic opening legs.
2. **Operational stability is solid** — business day, collection timing, deposit dates, and reconciliation should be stable in production first; fiscal close amplifies any drift or date bugs.
3. **At least one staging tenant** can run readiness with zero drift, no blocking deposits/cash-outs/recon exceptions, and realistic arrears/loan data.

15.2 Why Phase 1 now

- Surfaces whether close is even feasible for a tenant today (pool drift, open items, cycle gaps).
- Reuses existing services (`MemberInvariantService`, `MasterAccountInvariantService`, `MemberOpeningBalanceService`) without changing ledger data.
- Gives admins visibility before committing to purge semantics.

15.3 Why not full implementation yet

- **Purge is high-risk and irreversible** — wrong cut-off or deleted arrears/installment rows breaks collection and eligibility.
- **Cross-cutting hooks** — `books_closed_through` must be enforced everywhere money moves (`AccountingService`, contribution/EMI collection, backdated postings).
- **Compliance/audit** — export and archive strategy should be agreed before Tier B deletes anything.

15.4 Practical sequence

Now → Phase 1 (readiness UI + report)
Next → Confirm §13 policies with stakeholders
Then → Phase 2 on staging (snapshot + roll-forward, keep all history)
Only after → Phase 3/4 purge, with exports mandatory and invariant tests green

15.5 Bottom line

Treat full fiscal close as a **major accounting feature**, not a housekeeping task. If no tenant needs year-end close in the next few months, Phase 1 still pays off as a periodic health check; defer Phases 2–4 until there is a concrete close date and a signed-off retention policy.

Suggested first ticket: Fiscal calendar settings + `FiscalCloseReadinessService` + Filament report page (read-only).

16. Related documents

- [Fund management system requirements](#) — migration cut-off, opening balances
- [Collection cycle workflow](#) — phases and invariants
- [Fund flow implementation](#) — bank mirror and clearance
- Accounting master/member sync rule — `.cursor/rules/accounting-master-member-sync.mdc`